task1.py | task1_output.txt | task2.py | task2_output.txt | task3.py | task3_output.txt | task4.py

```python
import time
import threading
import queue
import os
TOTAL_ITEMS = 1000
WORKERS = 4
def process_item(i):  2 usages
    time.sleep(0.0001 * (i % 20))

def run_static():  1 usage
    chunk_size = TOTAL_ITEMS // WORKERS
    finish_times = [0.0] * WORKERS
    def worker(worker_id, start, end):
        for i in range(start, end):
            process_item(i)
        finish_times[worker_id] = time.perf_counter()
    threads = []
    start_time = time.perf_counter()

    for worker_id in range(WORKERS):
        start = worker_id * chunk_size
        end = start + chunk_size
        thread = threading.Thread(
            target=worker,
            args=(worker_id, start, end)
        )
        threads.append(thread)
        thread.start()
    for thread in threads:
        thread.join()

```

task1.py | task1_output.txt | task2.py | task2_output.txt | task3.py | task3_output.txt | task4.py

```python
def run_static():  1 usage
    total_time = time.perf_counter() - start_time
    completion_delta = max(finish_times) - min(finish_times)
    return total_time, completion_delta
def run_dynamic(chunk_size=1):  2 usages
    work_queue = queue.Queue()
    finish_times = [0.0] * WORKERS

    for start in range(0, TOTAL_ITEMS, chunk_size):
        end = min(start + chunk_size, TOTAL_ITEMS)
        work_queue.put((start, end))
    def worker(worker_id):
        while True:
            try:
                start, end = work_queue.get_nowait()
            except queue.Empty:
                break
            for i in range(start, end):
                process_item(i)
            work_queue.task_done()
        finish_times[worker_id] = time.perf_counter()
    threads = []
    start_time = time.perf_counter()

    for worker_id in range(WORKERS):
        thread = threading.Thread(target=worker, args=(worker_id,))
        threads.append(thread)
        thread.start()
    for thread in threads:
        thread.join()
```

task1.py | task1_output.txt | task2.py | task2_output.txt | task3.py | task3_output.txt | task4.py

```
35      def run_dynamic(chunk_size=1):  2 usages
57              threads.append(thread)
58              thread.start()
59          for thread in threads:
60              thread.join()
61
62          total_time = time.perf_counter() - start_time
63          completion_delta = max(finish_times) - min(finish_times)
64          return total_time, completion_delta
65
66      def main():  1 usage
67          print("TASK 4: WORK STEALING & SCHEDULING UNDER SKEW")
68          print(f"Operating System: {os.name}")
69          print(f"Total items: {TOTAL_ITEMS}")
70          print(f"Parallel workers: {WORKERS}")
71          print("Workload: sleep(0.0001 * (i % 20))")
72          print()
73          static_time, static_delta = run_static()
74          dynamic1_time, dynamic1_delta = run_dynamic(chunk_size=1)
75          dynamic5_time, dynamic5_delta = run_dynamic(chunk_size=5)
76          print(f"{'Strategy':<25}{'Total Time (s)':<20}{'Completion Delta (s)'}")
77          print("-" * 65)
78          print(f"{'Static chunks':<25}{static_time:<20.6f}{static_delta:.6f}")
79          print(f"{'Dynamic, chunk=1':<25}{dynamic1_time:<20.6f}{dynamic1_delta:.6f}")
80          print(f"{'Dynamic, chunk=5':<25}{dynamic5_time:<20.6f}{dynamic5_delta:.6f}")
81          print()
82          print("Completion delta = last thread finish time - first thread finish time.")
83          print("All three strategies process the same 1,000 items.")
84
85      if __name__ == "__main__":
86          main()
```

Run | task4

```
C:\Users\Акерке\.local\bin\uv.exe run C:/Users/Акерке/PycharmProjects/Pyt
TASK 4: WORK STEALING & SCHEDULING UNDER SKEW
Operating System: nt
Total items: 1000
Parallel workers: 4
Workload: sleep(0.0001 * (i % 20))

Strategy                  Total Time (s)      Completion Delta (s)
-----------------------------------------------------------------
Static chunks             0.341345            0.020515
Dynamic, chunk=1          0.337831            0.001764
Dynamic, chunk=5          0.325730            0.008730

Completion delta = last thread finish time - first thread finish time.
All three strategies process the same 1,000 items.

Process finished with exit code 0
```

task1.py | task1_output.txt | task2.py | task2_output.txt | task3.py | task3_output.txt | task4.py | task4_output.txt

```
TASK 4: WORK STEALING & SCHEDULING UNDER SKEW
Operating System: nt
Total items: 1000
Parallel workers: 4
Workload: sleep(0.0001 * (i % 20))

Strategy                Total Time (s)      Completion Delta (s)
----------------------------------------------------------------
Static chunks           0.341345            0.020515
Dynamic, chunk=1        0.337831            0.001764
Dynamic, chunk=5        0.325730            0.008730

Completion delta = last thread finish time - first thread finish time.
All three strategies process the same 1,000 items.
```